Project 01 — Campus Foundation

Building a Fully Segmented Enterprise Campus Network

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Platform: Cisco CML 2.9 | IOL + IOL-L2

Methodology: Build → Verify → Break → Fix

Setup and Scenario

Business Context

A growing technology company needs to segment its headquarters network into separate security zones for Engineering, Sales, Guest users, and Network Management. The current flat network allows all devices to communicate freely — a Guest laptop can reach Engineering servers, and management interfaces are reachable from any workstation. This lab designs and builds the complete segmentation from a written requirement.

What You Will Build

A multi-VLAN campus using two distribution switches (HQ-DSW1, HQ-DSW2), two access switches (HQ-ASW1, HQ-ASW2), and one router (HQ-RTR1) performing router-on-a-stick inter-VLAN routing. The design includes:

- Four production VLANs (100 Engineering, 200 Sales, 300 Guest, 999 Management) plus VLAN 1000 as a hardened native VLAN
- 802.1Q trunks with explicit allowed VLAN lists on all switch-to-switch links
- Router-on-a-stick subinterfaces as the default gateway for each VLAN
- Rapid-PVST+ with deliberate root bridge election across both distribution switches
- PortFast + BPDU Guard on all access ports
- SSH v2 management restricted to VLAN 999 only via ACL

STAR Summary

Situation: Studying VLANs, trunking, and inter-VLAN routing as separate topics without a real design exercise.

Task: Design and build a fully segmented campus network from a written requirement.

Action: Built multi-VLAN campus (100/200/300/999) with trunks, router-on-a-stick inter-VLAN routing, deliberate STP root election, and SSH-only management restricted to VLAN 999.

Result: Can design a multi-VLAN campus from scratch, explain every decision, and verify all technologies working together.

Technologies Used

- VLANs (100, 200, 300, 999, 1000)
- 802.1Q Trunking
- Native VLAN hardening (VLAN 1000)
- Router-on-a-Stick inter-VLAN routing
- Rapid-PVST+ Spanning Tree

- PortFast + BPDU Guard
- SSH v2 with VTY ACL (management VLAN only)
- CDP for neighbor discovery
- Interface descriptions

Network Topology

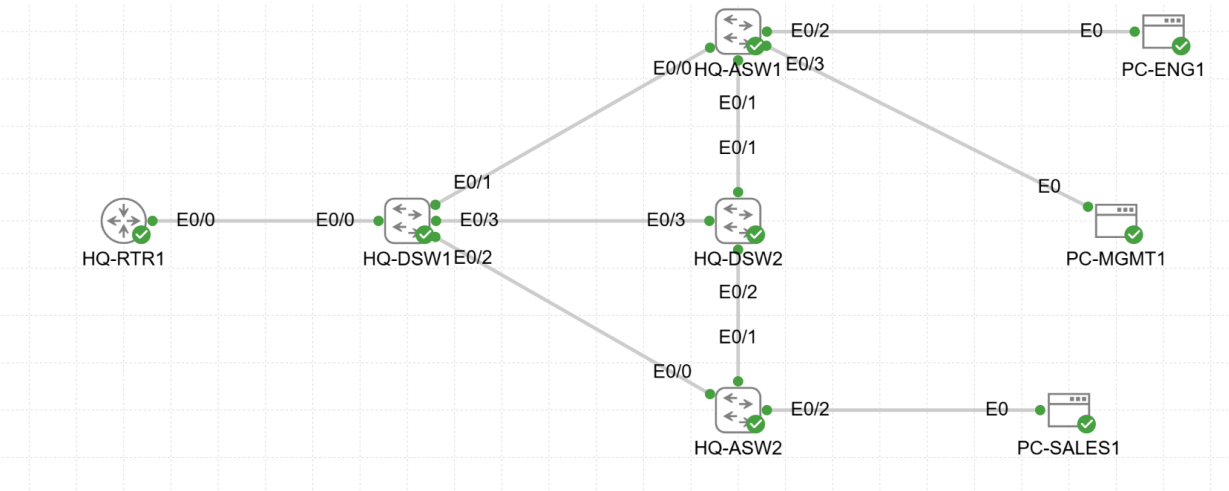

Figure T.1 — CML Topology — Project 01 Campus Foundation

IP Addressing

Device	Interface	IP Address	Subnet	Purpose
HQ-RTR1	E0/0.100	10.1.100.1	/24	Engineering gateway
HQ-RTR1	E0/0.200	10.1.200.1	/24	Sales gateway
HQ-RTR1	E0/0.300	10.1.44.1	/24	Guest gateway
HQ-RTR1	E0/0.999	10.1.99.1	/24	Management gateway
HQ-RTR1	Loopback0	10.0.255.1	/32	Router ID
HQ-DSW1	VLAN 999	10.1.99.11	/24	Management SVI
HQ-DSW2	VLAN 999	10.1.99.12	/24	Management SVI
HQ-ASW1	VLAN 999	10.1.99.13	/24	Management SVI
HQ-ASW2	VLAN 999	10.1.99.14	/24	Management SVI
PC-ENG1	eth0	10.1.100.10	/24	Engineering endpoint
PC-SALES1	eth0	10.1.200.10	/24	Sales endpoint
PC-MGMT1	eth0	10.1.99.100	/24	Management endpoint

VLAN	Name	Subnet	Purpose
100	Engineering	10.1.100.0/24	Developer workstations
200	Sales	10.1.200.0/24	Sales team
300	Guest	10.1.44.0/24	Guest isolation
999	Management	10.1.99.0/24	Network device management
1000	NATIVE-UNUSED	none	Native VLAN (no traffic)

Phase 1: VLAN Configuration

Goal: Create the VLAN database on all switches and configure access ports for end devices.

Step 1

Open the console for HQ-DSW1. Enter configuration mode and create the VLAN database:

```
enable
configure terminal
hostname HQ-DSW1
lldp run
vlan 100
 name Engineering
vlan 200
 name Sales
vlan 300
 name Guest
vlan 999
 name Management
vlan 1000
 name NATIVE-UNUSED
```

> **WHY:** *VLANs must exist in the VLAN database before they can be assigned to ports. VLAN 1000 is created as an unused native VLAN — a security hardening step to ensure untagged frames on trunks go nowhere.*

Step 2

Configure the Management SVI and default gateway on HQ-DSW1:

```
interface vlan 999
 ip address 10.1.99.11 255.255.255.0
 no shutdown
ip default-gateway 10.1.99.1
```

> **WHY:** *Without a Layer 3 SVI, the switch cannot be reached via SSH for management. The default gateway tells the switch where to forward packets destined beyond its directly connected subnets.*

Step 3

Repeat the VLAN database and SVI configuration on HQ-DSW2 (ip: 10.1.99.12), HQ-ASW1 (ip: 10.1.99.13), and HQ-ASW2 (ip: 10.1.99.14). Use the same VLAN names and the same management gateway 10.1.99.1.

Step 4

On HQ-ASW1, configure access ports for endpoints:

```
interface Ethernet0/2
 description ACCESS-PC-ENG1-VLAN100
```

```
 switchport mode access
 switchport access vlan 100
 no shutdown
interface Ethernet0/3
 description ACCESS-PC-MGMT1-VLAN999
 switchport mode access
 switchport access vlan 999
 no shutdown
```

> **WHY:** *switchport mode access restricts the port to a single VLAN and disables trunk negotiation. End devices do not understand 802.1Q tags — they need an untagged access port.*

Step 5

On HQ-ASW2, configure the access port for PC-SALES1 on interface Ethernet0/2, VLAN 200.

Verification

```
show vlan brief
```

Expected result: VLANs 100, 200, 300, 999, 1000 all show as "active". Ethernet0/2 listed under VLAN 100 on ASW1, Ethernet0/3 under VLAN 999.

```
HQ-ASW1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
100  Engineering                      active    Et0/2
200  Sales                            active
300  Guest                            active
999  Management                       active    Et0/3
1000 NATIVE-UNUSED                    active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

Figure 1.1 — show vlan brief — HQ-ASW1

Phase 2: 802.1Q Trunking

Goal: Configure 802.1Q trunk links between all switches so tagged VLAN traffic can cross switch-to-switch links.

Step 1

On HQ-DSW1, configure trunk links to all connected devices:

```
interface Ethernet0/0
 description TRUNK-TO-HQ-RTR1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999
 switchport nonegotiate
 no shutdown
interface Ethernet0/1
 description TRUNK-TO-HQ-ASW1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999
 switchport nonegotiate
 no shutdown
interface Ethernet0/2
 description TRUNK-TO-HQ-ASW2-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999
 switchport nonegotiate
 no shutdown
interface Ethernet0/3
 description TRUNK-TO-HQ-DSW2-E0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999
 switchport nonegotiate
 no shutdown
```

WHY: *switchport trunk encapsulation dot1q must be set BEFORE mode trunk on IOL-L2 — skipping this causes IOS to reject the trunk command. The explicit allowed VLAN list prevents unnecessary VLANs from traversing trunks. VLAN 1000 as native means untagged frames are discarded.*

Step 2

On HQ-DSW2, configure trunks on E0/1 (to ASW1), E0/2 (to ASW2), E0/3 (to DSW1) using the same trunk template above with correct descriptions.

Step 3

On HQ-ASW1, configure trunks on E0/0 (primary uplink to DSW1) and E0/1 (redundant uplink to DSW2).

Step 4

On HQ-ASW2, configure trunks on E0/0 (to DSW1) and E0/1 (to DSW2).

Verification

```
show interfaces trunk
show cdp neighbors
```

Expected: All trunk ports listed with Mode=on, Encapsulation=802.1q, Native VLAN=1000. Note: "Vlans in spanning tree forwarding state = none" on a redundant port is CORRECT — STP is blocking the loop path. This is not a failure.

```
HQ-DSW1#sh int trunk

Port        Mode         Encapsulation  Status       Native vlan
Et0/0       on           802.1q         trunking     1000
Et0/1       on           802.1q         trunking     1000
Et0/2       on           802.1q         trunking     1000
Et0/3       on           802.1q         trunking     1000

Port        Vlans allowed on trunk
Et0/0       100,200,300,999
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999

Port        Vlans allowed and active in management domain
Et0/0       100,200,300,999
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999

Port        Vlans in spanning tree forwarding state and not pruned
Et0/0       100,200,300,999
Et0/1       100,200,300,999
Et0/2       100,200,300,999

Port        Vlans in spanning tree forwarding state and not pruned
Et0/3       100,200,300,999
HQ-DSW1#
```

Figure 2.1 — show interfaces trunk — HQ-DSW1

Phase 3: Inter-VLAN Routing (Router-on-a-Stick)

Goal: Configure HQ-RTR1 subinterfaces so it can route traffic between all VLANs.

Step 1

On HQ-RTR1, bring up the physical trunk interface (no IP — the subinterfaces handle routing):

```
enable
configure terminal
hostname HQ-RTR1
lldp run
interface Ethernet0/0
 description TRUNK-TO-HQ-DSW1-E0/0
 no ip address
 no shutdown
```

> **WHY:** *The parent interface is a "pipe" only — it carries 802.1Q-tagged frames from the switch. No IP is assigned here because routing happens on the subinterfaces.*

Step 2

Create subinterfaces for each VLAN:

```
interface Ethernet0/0.100
 description GATEWAY-VLAN100-ENGINEERING
 encapsulation dot1Q 100
 ip address 10.1.100.1 255.255.255.0
interface Ethernet0/0.200
 description GATEWAY-VLAN200-SALES
 encapsulation dot1Q 200
 ip address 10.1.200.1 255.255.255.0
interface Ethernet0/0.300
 description GATEWAY-VLAN300-GUEST
 encapsulation dot1Q 300
 ip address 10.1.44.1 255.255.255.0
interface Ethernet0/0.999
 description GATEWAY-VLAN999-MANAGEMENT
 encapsulation dot1Q 999
 ip address 10.1.99.1 255.255.255.0
interface Loopback0
 description ROUTER-ID-AND-MGMT
 ip address 10.0.255.1 255.255.255.255
end
write memory
```

> **WHY:** *encapsulation dot1Q tells IOS which VLAN to strip/add tags for on this subinterface. Each subinterface becomes the default gateway for all hosts in that VLAN. The Loopback provides a stable router ID for OSPF in Project 03.*

Step 3

Set the domain name and generate RSA keys for SSH (also required for Phase 5):

```
configure terminal
ip domain-name lab.local
crypto key generate rsa general-keys modulus 2048
```

Step 4

Assign static IPs to endpoints. On PC-ENG1:

```
ip addr add 10.1.100.10/24 dev eth0
ip link set eth0 up
ip route add default via 10.1.100.1
```

On PC-SALES1 (10.1.200.10, gw 10.1.200.1) and PC-MGMT1 (10.1.99.100, gw 10.1.99.1) — same pattern.

Verification

```
show ip interface brief     ! on HQ-RTR1
show ip route               ! on HQ-RTR1
ping 10.1.200.10            ! from PC-ENG1 — cross-VLAN ping
```

Expected: All subinterfaces up/up with correct IPs. Routing table shows directly connected routes for all 4 VLANs. Cross-VLAN ping succeeds.

```
PC-MGMT1:~# ping 10.1.200.10
PING 10.1.200.10 (10.1.200.10): 56 data bytes
64 bytes from 10.1.200.10: seq=0 ttl=63 time=8.443 ms
64 bytes from 10.1.200.10: seq=1 ttl=63 time=8.197 ms
64 bytes from 10.1.200.10: seq=2 ttl=63 time=7.374 ms
64 bytes from 10.1.200.10: seq=3 ttl=63 time=8.096 ms
64 bytes from 10.1.200.10: seq=4 ttl=63 time=6.986 ms
64 bytes from 10.1.200.10: seq=5 ttl=63 time=7.809 ms
64 bytes from 10.1.200.10: seq=6 ttl=63 time=8.680 ms
64 bytes from 10.1.200.10: seq=7 ttl=63 time=7.356 ms
64 bytes from 10.1.200.10: seq=8 ttl=63 time=7.112 ms
64 bytes from 10.1.200.10: seq=9 ttl=63 time=7.457 ms
64 bytes from 10.1.200.10: seq=10 ttl=63 time=8.453 ms
64 bytes from 10.1.200.10: seq=11 ttl=63 time=7.902 ms
64 bytes from 10.1.200.10: seq=12 ttl=63 time=7.228 ms
```

Figure 3.1 — Cross-VLAN ping — PC-ENG1 to PC-SALES1

Phase 4: STP Hardening

Goal: Force a deliberate Spanning Tree root bridge election and enable PortFast + BPDU Guard on all access ports.

Step 1

On HQ-DSW1, set STP priorities to make it root for VLANs 100 and 999:

```
spanning-tree mode rapid-pvst
spanning-tree vlan 100 priority 4096
spanning-tree vlan 999 priority 4096
spanning-tree vlan 200 priority 8192
spanning-tree vlan 300 priority 8192
```

> **WHY:** *Default STP root election uses the lowest MAC address — unpredictable. Priority 4096 (1/8 of default 32768) guarantees DSW1 wins for VLANs 100 and 999. Rapid-PVST+ converges in 1–3 seconds vs 30–50 seconds for 802.1D.*

Step 2

On HQ-DSW2, set opposite priorities (root for VLANs 200/300, backup for 100/999):

```
spanning-tree mode rapid-pvst
spanning-tree vlan 200 priority 4096
spanning-tree vlan 300 priority 4096
spanning-tree vlan 100 priority 8192
spanning-tree vlan 999 priority 8192
```

> **WHY:** *Splitting root assignment across two distribution switches load-balances STP traffic. Each switch handles half the VLANs as primary root.*

Step 3

On HQ-ASW1 and HQ-ASW2, enable PortFast and BPDU Guard on access ports:

```
spanning-tree mode rapid-pvst
interface range Ethernet0/2 - 3
 spanning-tree portfast
 spanning-tree bpduguard enable
```

> **WHY:** *PortFast skips STP Listening and Learning states — PCs get connectivity immediately. BPDU Guard err-disables the port if a switch is plugged in, protecting the STP topology.*

Verification

```
show spanning-tree vlan 100     ! on HQ-DSW1 — look for "This bridge is the root"
```

```
show spanning-tree root          ! on HQ-ASW1 — verify correct root for each VLAN
```

```
HQ-DSW1#show spanning-tree vlan 100

VLAN0100
  Spanning tree enabled protocol rstp
  Root ID    Priority    4196
             Address     aabb.cc00.0a00
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4196   (priority 4096 sys-id-ext 100)
             Address     aabb.cc00.0a00
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -------------------------------
Et0/0               Desg FWD 100       128.1    P2p
Et0/1               Desg FWD 100       128.2    P2p
Et0/2               Desg FWD 100       128.3    P2p
Et0/3               Desg FWD 100       128.4    P2p


HQ-DSW1#
```

Figure 4.1 — STP Root confirmed — HQ-DSW1 VLAN 100

Phase 5: SSH Hardening

Goal: Restrict all device management to SSH v2, limited to the management VLAN only.

Step 1

Apply the following configuration to ALL five devices (HQ-RTR1, HQ-DSW1, HQ-DSW2, HQ-ASW1, HQ-ASW2):

```
ip domain-name lab.local
username admin privilege 15 secret CMLlab2025!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
line vty 0 4
 transport input ssh
 login local
 exec-timeout 10 0
 logging synchronous
 access-class 10 in
line con 0
 login local
 exec-timeout 10 0
 logging synchronous
access-list 10 permit 10.1.99.0 0.0.0.255
access-list 10 deny any log
no ip http server
no ip http secure-server
banner motd ^
===========================================
AUTHORIZED ACCESS ONLY
All access is logged and monitored
===========================================
^
service password-encryption
enable secret CMLenableP@ss!
end
write memory
```

> **WHY:** *transport input ssh: Telnet sends credentials in cleartext — never acceptable in production. access-class 10: Restricts SSH to 10.1.99.0/24 (management VLAN only). service password-encryption: Encrypts all cleartext passwords in running config.*

Step 2

Generate RSA keys on each device after the config above:

```
configure terminal
ip domain-name lab.local
crypto key generate rsa general-keys modulus 2048
```

NOTE: ip domain-name MUST be set before generating keys. Without keys, SSH port 22 never opens — show ip ssh will show "SSH Disabled".

Verification

```
show ip ssh                     ! should show: SSH Enabled - version 2.0
ssh admin@10.1.99.11            ! from PC-MGMT1 — should succeed
ssh admin@10.1.99.11            ! from PC-ENG1  — should be refused by ACL
```

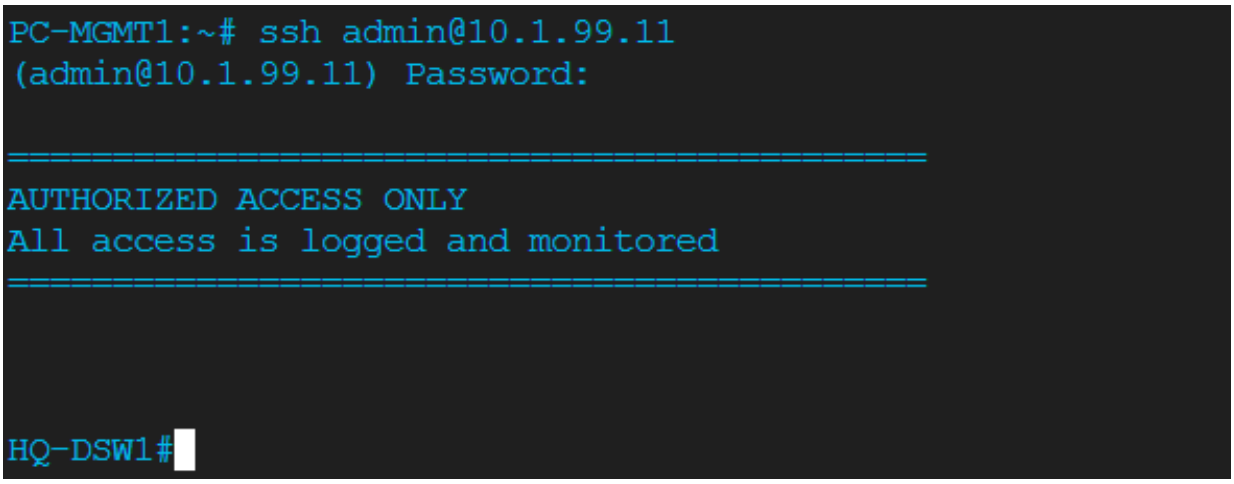

Figure 5.1 — SSH login success from PC-MGMT1

Final Verification Checklist

#	Test	Command	Device	Expected
1	VLANs exist	show vlan brief	All switches	100,200,300,999,1000 active
2	Trunks up	show interfaces trunk	DSW1	All uplinks trunking, native 1000
3	Subinterfaces	show ip interface brief	HQ-RTR1	E0/0.100–.999 all up/up
4	Cross-VLAN ping	ping 10.1.200.10	PC-ENG1	5/5 success
5	STP root DSW1	show spanning-tree vlan 100	HQ-DSW1	This bridge is the root
6	STP root DSW2	show spanning-tree vlan 200	HQ-DSW2	This bridge is the root
7	SSH from Mgmt	ssh admin@10.1.99.11	PC-MGMT1	Login succeeds
8	SSH blocked	ssh admin@10.1.99.11	PC-ENG1	Connection refused

Troubleshooting — Top 5 Issues

Issue 1 — Interface Naming Mismatch

Symptom: Commands targeting GigabitEthernet0/0 returned "Invalid interface" errors on every device.

Root Cause: CML IOL images use Ethernet0/x naming, not GigabitEthernet. IOL-L2 switches have one 4-port slot (E0/0–E0/3), not dual slots.

Fix: Ran show ip interface brief on every node after boot and adapted all configs to the actual interface names shown.

Issue 2 — Native VLAN Mismatch CDP Warning

Symptom: CDP warnings "NATIVE_VLAN_MISMATCH on Ethernet0/3 (1000) with HQ-DSW2 (1)" appeared on DSW1 console during build.

Root Cause: DSW1 was configured with native VLAN 1000 first; the far-end switches still had default native VLAN 1 while being configured.

Fix: Completed trunk configuration on all remaining switches. Warnings cleared automatically once both ends matched.

Issue 3 — STP Redundant Trunk Showing "none" for Forwarding VLANs

Symptom: show interfaces trunk on HQ-DSW2 showed "Vlans in spanning tree forwarding state = none" on redundant uplink ports.

Root Cause: Not a failure — Rapid-PVST+ was blocking the redundant path to prevent a Layer 2 loop. This is correct behavior in a dual-distribution design.

Fix: No fix required. Confirmed correct STP root layout and verified traffic was flowing via the primary path.

Issue 4 — SSH Connection Refused Despite VTY Config Applied

Symptom: SSH from PC-MGMT1 to HQ-DSW1 failed with "connection refused" even though ping succeeded.

Root Cause: crypto key generate rsa was not run after ip domain-name was set. Without RSA host keys, SSH port 22 never opens.

Fix: Re-ran ip domain-name lab.local followed by crypto key generate rsa general-keys modulus 2048 on all devices. IOS logged: %SSH-5-ENABLED: SSH 2.0 has been enabled.

Issue 5 — Break/Fix: VLAN 100 Removed from Trunk Allowed List

Symptom: PC-ENG1 lost all connectivity. Trunks remained up with no physical errors — no interface down messages.

Root Cause: Both ASW1 uplinks had VLAN 100 removed from the allowed VLAN list. The trunk stayed UP — the only evidence was in show interfaces trunk.

Fix: Added VLAN 100 back to the allowed list on both ASW1 uplinks with switchport trunk allowed vlan 100,200,300,999.

Break/Fix Challenge

Scenario: PC-ENG1 has lost all network connectivity. Trunks are up, no interfaces are down. Diagnose and fix using show commands only.

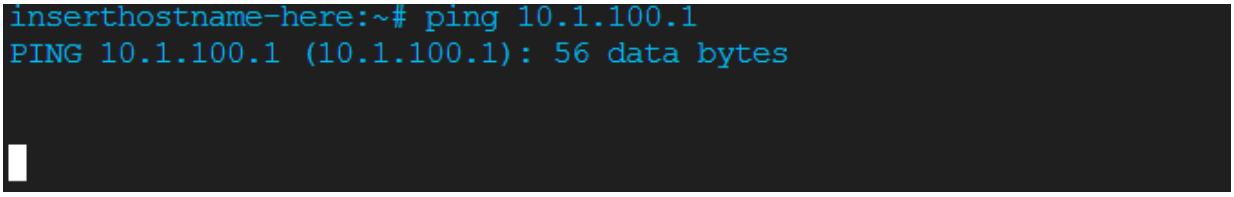

Figure B.1 — PC-ENG1 ping failing — VLAN 100 blocked

Step 1

Verify the problem from PC-ENG1:

```
ping 10.1.100.1     ! gateway — fails
ping 10.1.99.11     ! management switch — fails
```

Step 2

Check Layer 1/2 at HQ-ASW1:

```
show vlan brief
show interfaces trunk
```

Look for VLAN 100 missing from "Vlans allowed on trunk" column on E0/0 and E0/1.

Step 3

Apply fix on HQ-ASW1:

```
configure terminal
interface Ethernet0/0
 switchport trunk allowed vlan 100,200,300,999
interface Ethernet0/1
 switchport trunk allowed vlan 100,200,300,999
end
write memory
```

Step 4

Re-verify from PC-ENG1:

```
ping 10.1.200.10    ! cross-VLAN — should now succeed
```

```
inserthostname-here:~# ping 10.1.100.1
PING 10.1.100.1 (10.1.100.1): 56 data bytes


64 bytes from 10.1.100.1: seq=77 ttl=255 time=6.733 ms
64 bytes from 10.1.100.1: seq=78 ttl=255 time=4.287 ms
64 bytes from 10.1.100.1: seq=79 ttl=255 time=4.040 ms
64 bytes from 10.1.100.1: seq=80 ttl=255 time=3.483 ms
64 bytes from 10.1.100.1: seq=81 ttl=255 time=3.840 ms
64 bytes from 10.1.100.1: seq=82 ttl=255 time=4.194 ms
64 bytes from 10.1.100.1: seq=83 ttl=255 time=3.816 ms
64 bytes from 10.1.100.1: seq=84 ttl=255 time=3.581 ms
64 bytes from 10.1.100.1: seq=85 ttl=255 time=3.758 ms
64 bytes from 10.1.100.1: seq=86 ttl=255 time=4.531 ms
64 bytes from 10.1.100.1: seq=87 ttl=255 time=4.307 ms
```

Figure B.2 — PC-ENG1 connectivity restored

NOTE: Key Lesson: A wrong allowed VLAN list is one of the most common trunking failures. The trunk stays UP with no physical errors. The only diagnostic tool is show interfaces trunk — check the Vlans allowed column carefully.